

5.2 Solving Equations and Rearranging Formulas

Lesson Introduction

 Benchmarks	MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.		 Learning Targets	- I can solve multi-step, linear equations. - I can isolate a quantity of interest in a formula by rearranging a formula.
 Benchmark Coherence	Building On			Working Toward
	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context where all terms are rational numbers.			N/A
MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.				
 Essential Questions	- How can I write and solve an equation from a real-world context? - What is the relationship between solving equations and rearranging formulas?			
 Lesson Narrative	In this lesson, students extend their learning of multi-step linear equations to include equations with letters as the coefficients. Students will begin by completing an error analysis activity to reinforce what they have learned in the prior lesson with solving multi-step equations. They will then learn about how to apply the process of solving to equations where the coefficients are represented by letters.			
 Component Summary	5.2.1 Warm-Up (~5 min.)	Closest to zero puzzle (SE p. 236)	5.2.3 Guided Instruction (15-20 min.)	Introduction to rearranging formulas (SE p. 237)
	5.2.2 Collaboration: (10-15 min.)	Error analysis solving equations activity (SE p. 236)	5.2.4 Your Turn! (10 min)	Independent practice with rearranging formulas (SE p. 239)
	5.5.5 Wrap-Up (~5 min)	Growth Mindset activity (SE p. 239)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A		N/A	N/A
<u>Additional Terms:</u> N/A				

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not consider factoring if necessary to rearranging a formula. - Students may not see connection between solving equations and rearranging formulas. - Students may combine terms together that cannot be combined.	Students often struggle with rearranging formulas because of its abstract nature. Consider highlighting the process of rearranging formulas to what they have learned about solving equations in the prior lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 5.2.3 Guided Instruction introduces students to the concept of rearranging formulas. This is a concept students often misunderstand. Leveraging this instructional routine allows teachers to demonstrate thinking out loud and prompting students to reflect and respond.	MA.K12.MTR.4.1: Discussion and Reflection 5.2.2 Collaboration provides students with an opportunity to engage in an error analysis activity where they will determine whether an error existed. Students will recognize the error and provide the location of where the error occurred and how to correct it.
 Differentiate Instruction	“Why would we ever want to rearrange the variables? What is the point of this?” Learning happens when students are engaged and see meaning in their work. Rearranging formulas may seem like busy work without meaningful context. Consider launching the lesson with an exploration of the similarities and differences between the conversion formulas for Celsius and Fahrenheit. Why are they two different formulas? Who would use which formula and when? Are there other times in which it might be helpful to rearrange a formula?	
Lesson Notes:		

5.2.1 Warm-Up: Closest to Zero

Component Overview: Students will create an equation that is true using digit limitations.



5.2.1 Warm-Up: Closest to Zero

1. Place a digit in each blue box to make the equation true for only one value of x . Use only the digits 1 to 9, but digits can be used more than once.

$$\boxed{}x + \boxed{} = \boxed{}x + \boxed{}$$

Answers will vary. $2x + 4 = 4x + 8$



Consider leveraging a Think-Pair-Share instructional routine with this activity. Allow students individual think time to try and determine their own method for finding an answer. Then allow them time to work with a partner to share their methods and answers.

5.2.2 Collaboration: Find the Error

Component Overview: Students will examine four completed equations to determine if an error occurred. For this activity students will need a partner.



5.2.2 Collaboration: Find the Error

Sandra, Jerry, Marly, and Xavier solved the same equation and each found different solutions. Each student's work and solution are shown in the table.

1. Work with your partner to examine the work shown and determine who solved the equation correctly. Then, for the students who made errors, circle the step where the error was made and explain the error.

Name	Student's Work and Solution	Is the solution correct?
Sandra	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $\frac{3}{4}(16 - 2w) = \frac{1}{3}w + 2$ $12 - \frac{3}{2}w = \frac{1}{3}w + 2$ $10 - \frac{3}{2}w = \frac{1}{3}w$ $10 = \frac{11}{6}w$ $\frac{60}{11} = w$	☐ Yes ☒ No
	Error Explanation	
	<i>Sandra removed the negative sign when rewriting the next step.</i>	
Jerry	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 = \frac{7}{6}w + 11$ $-14 = \frac{7}{6}w$ $-12 = w$	☐ Yes ☒ No
	Error Explanation	
	<i>When subtracting $\frac{3}{2}w$ from $\frac{7}{6}w$, it should have been negative, not positive.</i>	



5.2.2 Collaboration intentionally targets common misconceptions. Take note of each corresponding mistake and incorporate each as a learning opportunity when students themselves make the same mistake – “How is your work the same or different than what Sandra did?”

- Sandra: Associated the negative sign with the leading integer rather than the coefficient of the quantity in parentheses
- Jerry: Operations with fractions
- Marly: No errors
- Xavier: Multiplying two negative values using the Distributive Property

5.2.2 Collaboration: Find the Error (Continued)

Marly	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 + \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 + \frac{3}{2}w = \frac{1}{3}w + 11$ $\frac{3}{2}w = \frac{1}{3}w + 14$ $\frac{7}{6}w = 14$ $w = 12$	☐ Yes ☐ No
	Error Explanation <i>n/a</i>	
Xavier	$9 - \frac{3}{4}(16 - 2w) = \frac{1}{3}w + 11$ $9 - 12 - \frac{3}{2}w = \frac{1}{3}w + 11$ $-3 - \frac{3}{2}w = \frac{1}{3}w + 11$ $-\frac{3}{2}w = \frac{1}{3}w + 14$ $-\frac{5}{6}w = 14$ $w = -16.8$	☐ Yes ☐ No
	Error Explanation <i>Xavier made an error with distribution with his integer operations.</i>	



How can you confirm whether the solution is the actual solution to the equation?



The error for Xavier with the Distributive Property is a very common error that students make. Use this an example for a classroom discussion about the error and ways to ensure that the final answer is correct.

5.2.3 Guided Instruction: Rearranging Formulas

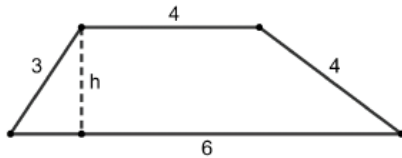
Component Overview: Students will be introduced to rearranging formulas using three different problems.



5.2.3 Guided Instruction: Rearranging Formulas

The formula for the area of a trapezoid is $A = \frac{1}{2}(b_1 + b_2)h$, where b_1 and b_2 are the lengths of the bases and h is the height of the trapezoid.

Stacey knows the area of the trapezoid shown is 12.5 square units, and uses the values given to determine the height, h , of the trapezoid. Her work is shown.



- Complete the table to explain each step of Stacey's work.

Stacey's Work	Explanation of Steps
$A = \frac{1}{2}(b_1 + b_2)h$	Formula for area of a trapezoid
$12.5 = \frac{1}{2}(4 + 6)h$	Substitute known values
$25 = (4 + 6)h$	<i>Multiplication Property of Equality</i>
$\frac{25}{(4 + 6)} = h$	<i>Division Property of Equality</i>
$\frac{25}{(10)} = h$	<i>Addition</i>
$2.5 = h$	<i>Division</i>



In this activity, students will learn about how to apply the process of solving to equations where the coefficients are represented by letters.

Students have already become familiar with solving multi-step equations. Connect that process and the steps they are familiar with to the concept of rearranging formulas so that students can understand the connection.

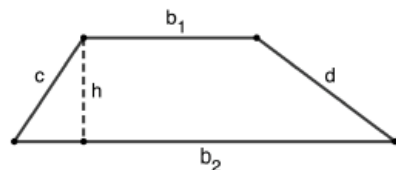


Consider modifying the steps to completing question 1. If students are struggling, guide them through the steps of Stacey's work from start to finish so that they are clear on the steps to solving. Then go back and justify each step so that students are hearing the process twice.

5.2.3 Guided Instruction: Rearranging Formulas (Continued.)

2. Stacey wants to rearrange the area formula for a trapezoid to isolate the variable h before replacing the values of the other variables.

- a. Use Stacey's work on the previous problem to help solve the equation for h .



$$A = \frac{1}{2}(b_1 + b_2)h$$

$$2A = (b_1 + b_2)h$$

$$\frac{2A}{(b_1 + b_2)} = h$$

- b. Check your work by substituting the values from the previous question. If correct, $h = 2.5$.

$$\frac{2(12.5)}{4 + 6} = 2.5$$

$$\frac{25}{10} = 2.5$$

$$2.5 = 2.5$$

- c. Explain the similarities and differences between rearranging the formula to isolate h and Stacey's work to find the value of h .

Similarities	Differences
Answers will vary. The steps were the same.	Answers will vary. One used variables only and the others used numeric values.



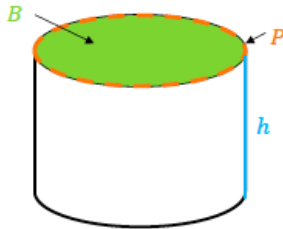
Consider releasing students to work on question 2 with a partner. Circulate the room, supporting students as they work through the problem. When time is called, review each part of the question and the correct answer to ensure there is a consensus on the correct answer before moving forward.



What was the most challenging part of rearranging this formula?

5.2.3 Guided Instruction: Rearranging Formulas (Continued.)

3. The formula for the total surface area of a cylinder, S , is $S = 2B + Ph$, where B is the area of the base, P is the perimeter of the base, and h is the height of the cylinder.



Release students to try question 3 on their own. Have them check their answer and their work with a partner before reviewing together as a whole class.

Rearrange the formula for the surface area of a cylinder to isolate the variable h .

$$\begin{aligned} S &= 2B + Ph \\ S - 2B &= Ph \\ \frac{S - 2B}{P} &= h \end{aligned}$$

5.2.4 Your Turn!

Component Overview: Students will complete three independent practice questions on rearranging formulas.



5.2.4 Your Turn!

Rearrange the formulas to isolate the specified variables.

1. The formula for force, F , is mass, m , times acceleration, a , or $F = ma$. Solve for mass, m .

$$\frac{F}{a} = m$$

2. The formula $C = 2\pi r$ is used to find the circumference, C , of a circle with radius r . Solve the equation for the radius, r .

$$\frac{C}{2\pi} = r$$

3. The formula to convert degrees Fahrenheit, F , into degrees Celsius, C , is $C = \frac{5}{9}(F - 32)$. Solve for F .

$$\frac{9}{5}C = F - 32$$

$$\frac{9}{5}C + 32 = F$$



For students who may be struggling, consider providing the “read aloud” approach. Have students read the original equations.

For example, question 1 would read “ F is equal to m times a ”. To solve for m , complete the process backwards using inverse operations. Instead of “times”, use “multiply”. Some students make better sense of mathematics by verbally processing the information.

5.2.5 Wrap-Up: Growth Mindset

Component Overview: Students will complete a Growth Mindset reflection activity answering three questions.



5.2.5 Wrap-Up: Growth Mindset

1. Choose **two** of the following prompts to complete.
 - a. Describe a new strategy you learned today.
Answers will vary.
 - b. Describe a mistake you made and what you learned from it.
Answers will vary.
 - c. Explain one way you challenged yourself today.
Answers will vary.



Consider tracking student responses to revisit later in the unit. For example, students can compare their mistake from today with their responses to the wrap-up in Lesson 3 which also unpacks mistakes as learning opportunities.